

Column height: inspections per ZIP code. Color: share ending B or C.

CS 667 MIDTERM · TECHNICAL PRESENTATION

A bad grade is partly visible before the inspector arrives

Predicting restaurant inspection grades in New York City, using only what is known before the visit.

Rafiul Haider · Shadman Rahman

CS 667 Practical Data Science · Fall 2026 · Dr. Krishna Bathula, Pace University · October 2026

ACT I · THE SETUP

The prediction is only allowed to use what exists before the visit

Known beforehand

- Borough and location
- Cuisine
- Visit type (cycle, re-inspection, ...)
- The restaurant's own past inspections

→

Model

trained on earlier years, tested on later ones

→

A priority list

- **For DOHMH:** which visits and follow-ups to schedule first
- **For operators:** which risks to fix before the inspector arrives

NEVER USED Anything from the inspection being predicted: **its score, its violations, its grade date.** A model that reads the answer from the same visit has learned nothing anyone can act on.

Target: B or C = 1 vs A = 0

ACT I · THE SETUP

295,356 rows are not 295,356 inspections

295,356 rows

one row per **violation**, as NYC Open Data publishes it

96,897 inspections

grouped into one record per inspection

31,443 restaurants

each inspected one or more times

The grouping key

restaurant + **date** + **inspection type**

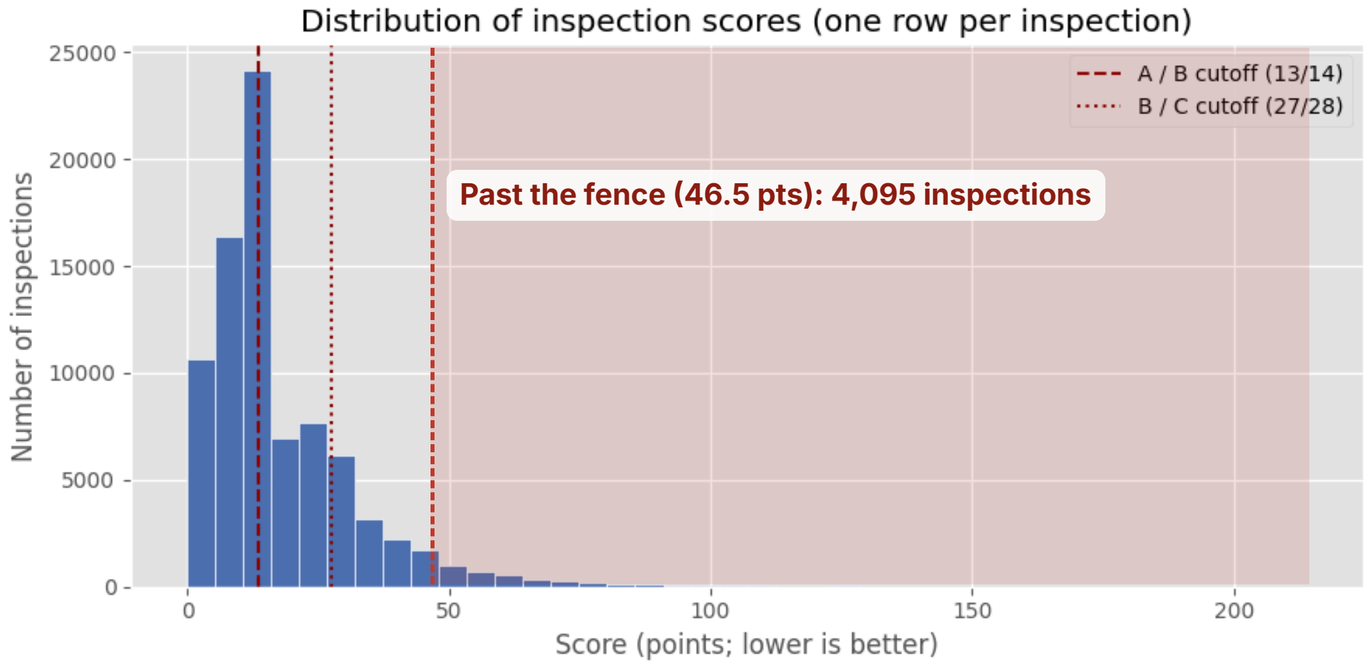
Score and grade repeat on every violation row, so any average taken at row level counts a nine-violation visit nine times.

Date alone is not enough: some restaurants have two different inspection types on the same day, and those would merge into one fake inspection.

Source: DOHMH New York City Restaurant Inspection Results, NYC Open Data (dataset 43nn-pn8j), raw file.

ACT II · THE SURFACE LIES

The median inspection passes — the tail is the story



Inspection score, one bar per score band (notebook figure). **Lower is better**: 0 to 13 points is an A. The shaded band is past the outlier fence.

13

median score, exactly the top of the A band

17.9

mean score, dragged up by the tail

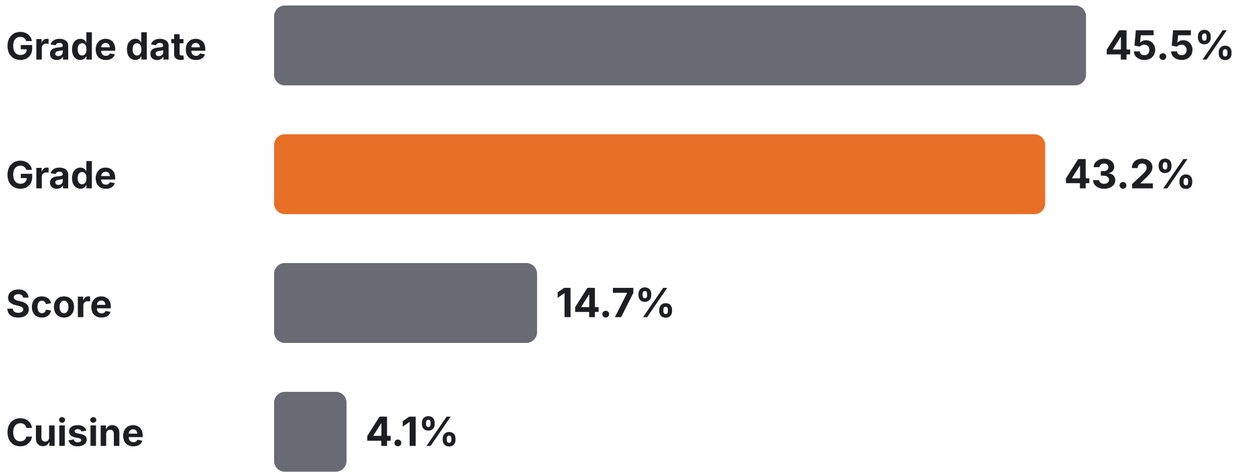
4,095

inspections past the fence at **46.5** points; the worst scored **214**

ACT II · THE SURFACE LIES

43% of inspections have no grade, and that is structure, not dirt

Share of inspections with the field missing



A missing grade is mostly a **pending** grade: the inspection happened, and the letter is not settled yet.

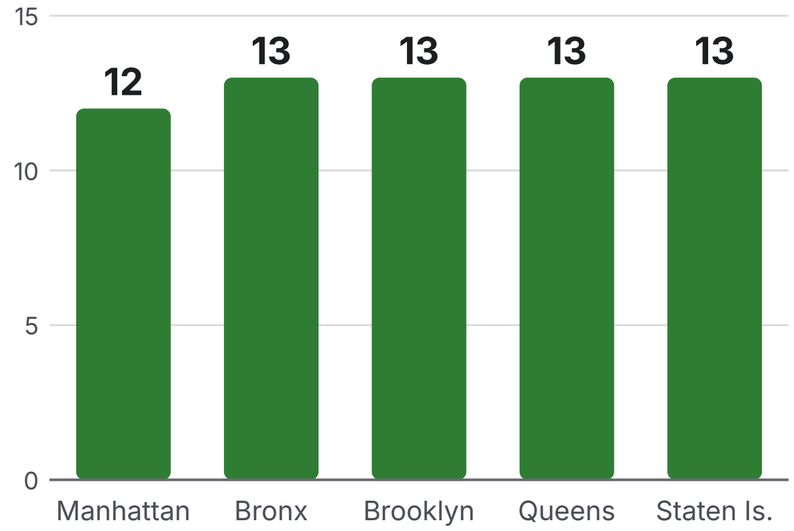
The interim codes N, Z and P sit in the grade column for the same reason. The grading process has a timeline, and the file is a snapshot of it.

So we treat a missing grade as information about where that inspection sits in the timeline, and we don't fill it in.

ACT II · THE SURFACE LIES

Borough tells you almost nothing; cuisine tells you a little

Median score by borough



Points, **lower is better**. Bar color shows the grade that median would earn. 84 scored inspections with no borough recorded are not shown.

Median score, the most-inspected cuisines (click one)



Points, **lower is better**. B or C share: among modeled inspections (graded, first visits excluded).

ACT III · THE PLACE IS NOT THE RISK

The strongest correlation in the file is partly mechanical — and one is fake



Trap 1 · score ↔ critical violations, 0.86

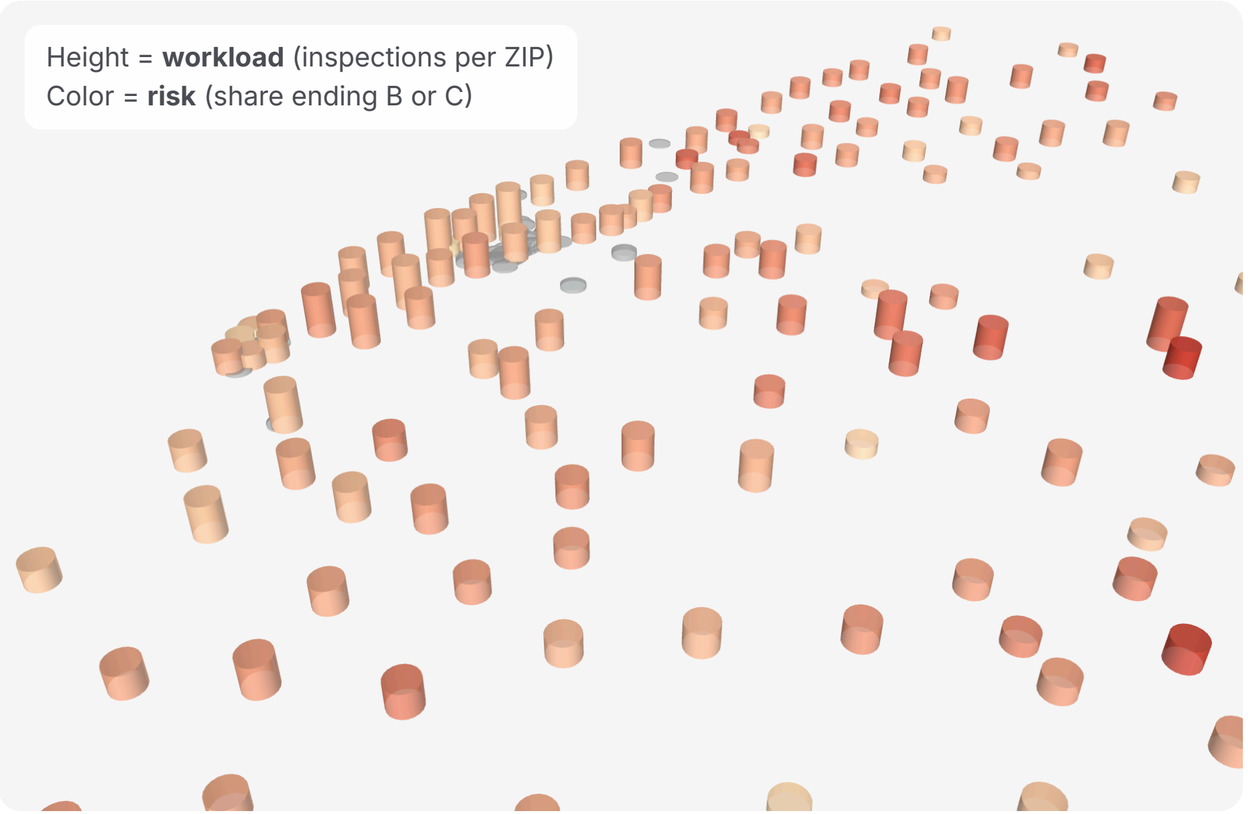
The score is **built from** violation points, so this correlation is arithmetic. It is also the reason same-inspection fields are banned as predictors.

Trap 2 · latitude ↔ longitude, −1.00

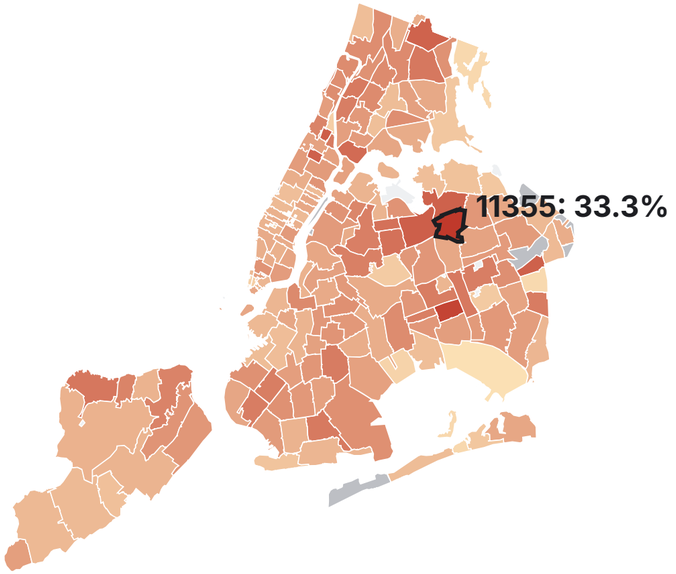
The city runs southwest to northeast, so as one goes up the other comes down. That is the shape of the map, and it tells us nothing about restaurants.

ACT III · THE PLACE IS NOT THE RISK

The busiest places are not the riskiest places



B or C share by ZIP code, flat view



0% → 30%+ B or C

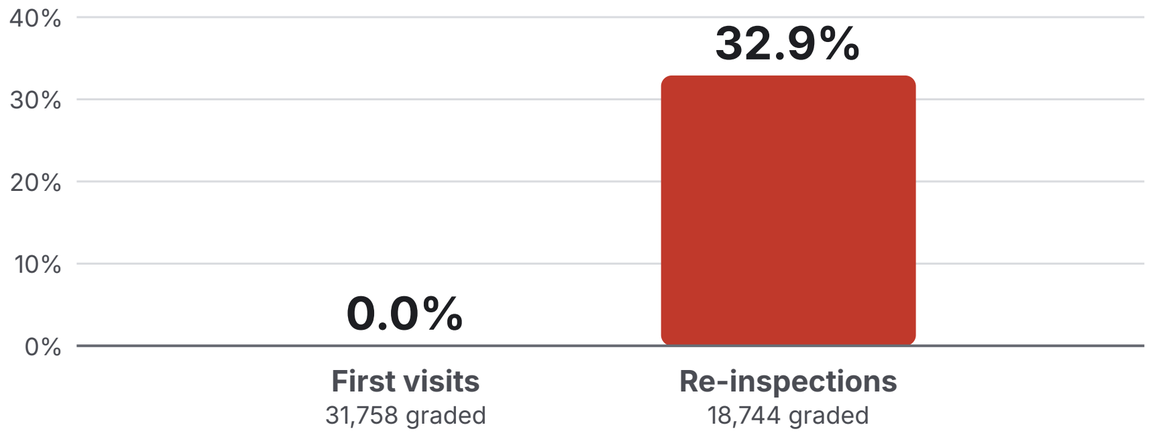
not enough inspections to judge (< 30 graded)

Worst reliable ZIP **11355**: **33.3%** of **342** graded. Citywide: **12.9%**.

ACT IV · THE DISCOVERY

First visits never fail — the grade is settled at the second visit

Share of graded inspections ending B or C



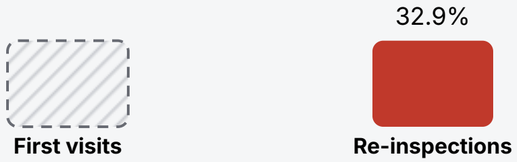
0 B or C grades on a first visit, out of **31,758** graded

Graded inspections by visit kind

	A	B or C
Initial inspection	31,758	0
All other inspections	12,745	6,497

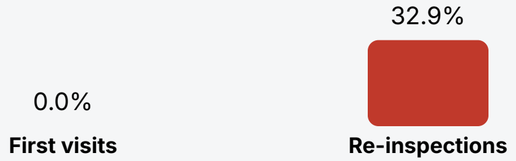
32.9% of graded re-inspections end B or C. A bad first visit leaves with the grade pending.

Before the reveal: what we expected



After: what the file says

Grade B or C | Scored 14+

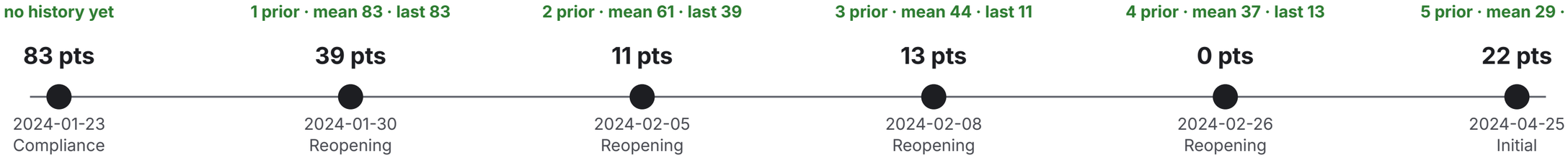


ACT V · THE HONEST MODEL

A restaurant's past is the fairest predictor of its next visit

One restaurant (camis 50098804), inspection by inspection · score in points, **lower is better**

Green: what the model can see before each visit



known before this visit

prior count
earlier inspections

known before this visit

prior mean score
average of earlier scores

known before this visit

prior B/C share
earlier grades that were B or C

known before this visit

prior mean critical
earlier critical violations

known before this visit

days since prior
gap since the last visit

known before this visit

last score
the most recent score

known before this visit

has history
any earlier visit at all

ZIP code
left out on purpose: borough + coordinates carry location

Train · up to the end of 2024 · 9,111 | **Test · 2025 onward · 9,887**

ACT V · THE HONEST MODEL

The pipeline is plain on purpose — and PCA told us to keep it that way

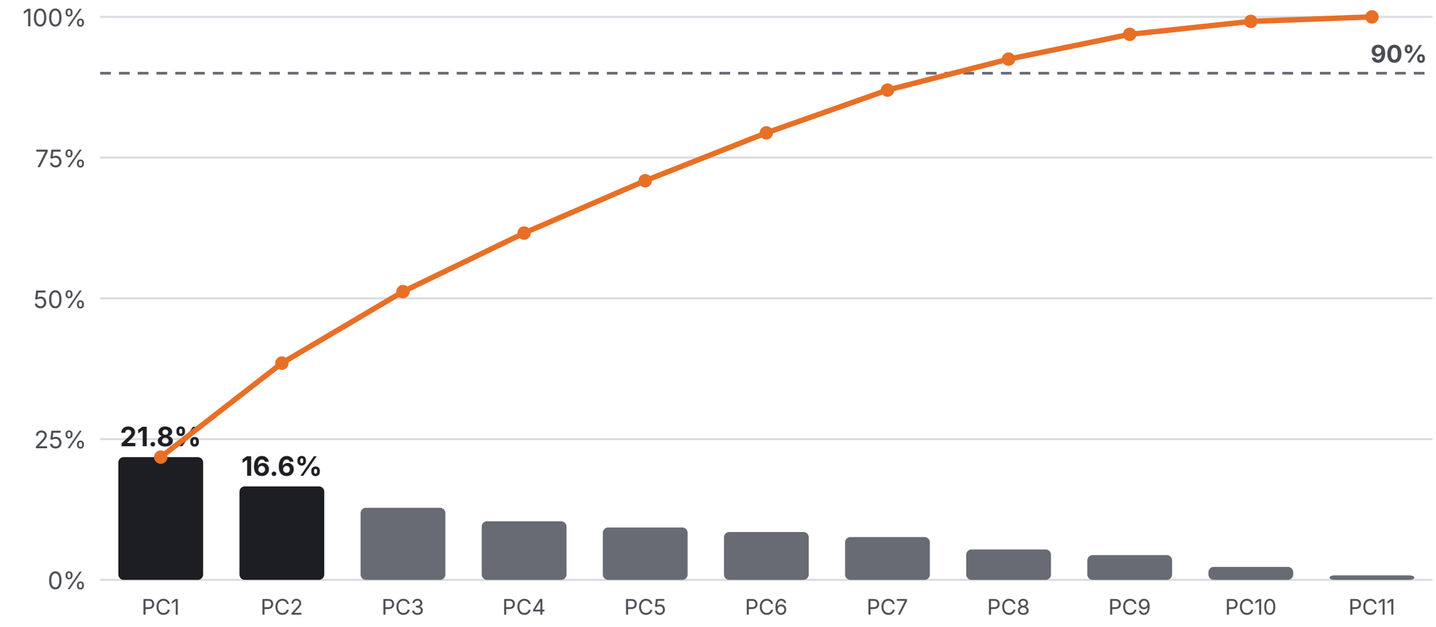
1. **One-hot encode**
   borough, grouped cuisine, inspection type

2. **Median imputation**
   a restaurant with no history gets the typical value

3. **Scaling**
   fitted on the training years, applied unchanged to the test years

Identical treatment for train and test: nothing is fitted on the test period.

PCA on the 11 numeric features: variance per component

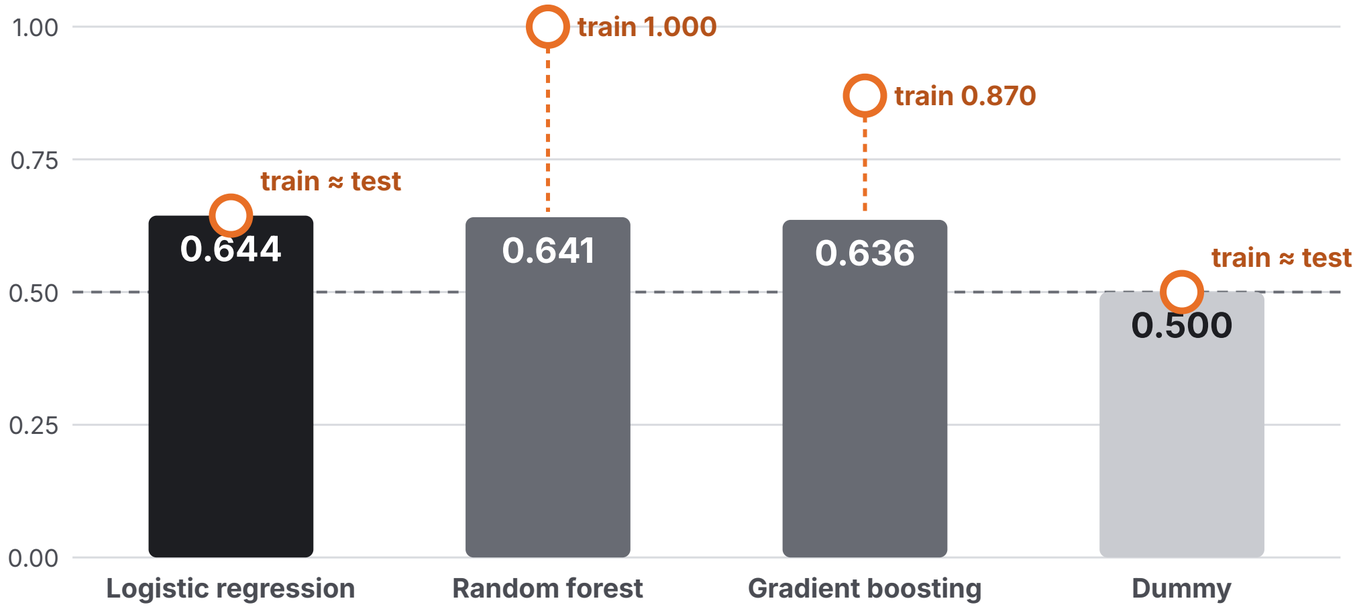


PC1 (21.8%) is a past-performance axis; PC2 (16.6%) a history-depth axis. 8 of 11 components for 90%: the features are not copies of each other.

ACT V · THE HONEST MODEL

The simplest model won — the forest memorized the training years

ROC-AUC: bars are the test years (2025 on), rings are the training years



Read the **gap**, not the bar: a ring far above its bar is a model that memorized. Dashed line: a coin flip.

THE DUMMY MODEL

~~66.8% accurate~~

Caught: 0 of 3,284 bad inspections.

It predicted A for everything. That is why accuracy is not the measure here.

ACT V · THE HONEST MODEL

The threshold is a policy choice, not a setting

Threshold

0.3 — 0.4 — **0.45 ★** — 0.5 — 0.6 — 0.7

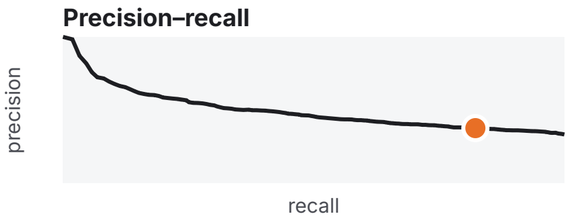
82.2%
recall: B or C inspections caught

37.6%
precision: flags that end B or C

0.516
F1

Catches ~82 of every 100 B or C inspections; about 6 in 10 flags still end A.



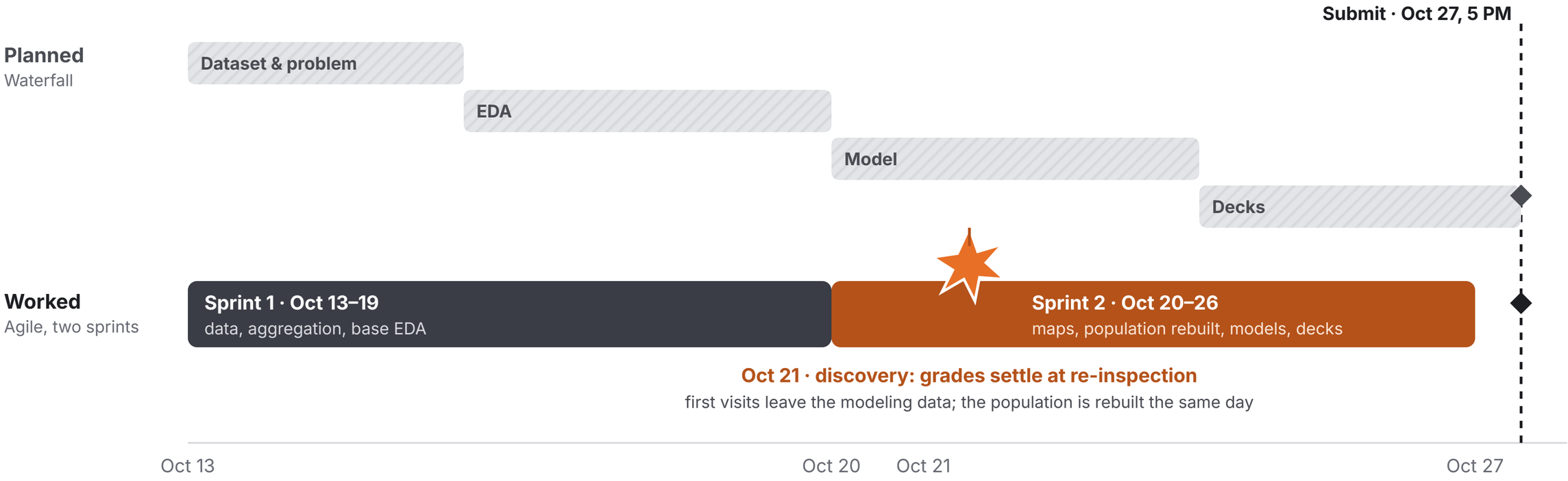


Test years (2025 on), 9,887 inspections

	Predicted A	Predicted B or C (flagged)
Ended A	**2,114** correctly left alone	**4,489** flagged, ended A (a wasted visit)
Ended B or C	**583** missed bad inspections	**2,701** caught bad inspections

ACT VI · THE PROCESS AND THE POINT

We planned like Waterfall and worked like Agile



The deadline never moved. The plan underneath it did, once, on evidence.

ACT VI · THE PROCESS AND THE POINT

A prioritization tool, not a judge

0.644

ranking quality (test ROC-AUC, 2025 on)

82.2%

of bad inspections caught at the chosen threshold, 0.45

33.3%

worst reliable ZIP's B or C share (11355), vs 12.9% citywide

What it is

- It orders a visit list from pre-visit information alone.
- It does not label a restaurant. A flag is a reason to look first.

Next steps

1. Richer history features
2. Per-visit-type models
3. Probability calibration

Questions.

The .pptx is the static record of this presentation; the HTML deck is the live version, with the animations, the threshold slider and the 3D map.

BACKUP B1 · Q&A

Risk by ZIP code holds up only where the counts do

Highest B or C share, reliable ZIPs (≥ 30 graded)

ZIP	Borough	Inspections	Graded	B or C
11355	Queens	822	342	33.3%
11419	Queens	449	203	28.1%
10030	Manhattan	93	46	23.9%
11368	Queens	1,023	499	23.6%
11428	Queens	152	77	23.4%
10026	Manhattan	216	103	23.3%
11354	Queens	1,844	793	23.2%
10466	Bronx	331	187	23.0%
10454	Bronx	342	162	21.6%
11372	Queens	1,481	678	20.6%

Lowest B or C share, reliable ZIPs

ZIP	Inspections	Graded	B or C
10281	57	39	0.0%
10121	189	163	1.2%
11430	136	103	1.9%
11429	98	64	3.1%
10464	93	61	3.3%
11693	92	60	3.3%
11239	80	48	4.2%
10006	167	104	4.8%
10119	79	42	4.8%
10280	76	42	4.8%

Citywide: 12.9% of graded inspections end B or C · 178 of 223 ZIPs reliable | **Data as of October 21, 2026**

BACKUP B2 · Q&A

At the default 0.5 threshold the model misses more bad inspections

	Predicted A	Predicted B or C
Ended A	**3,137** correctly left alone	**3,466** flagged, ended A
Ended B or C	**954** missed	**2,330** caught

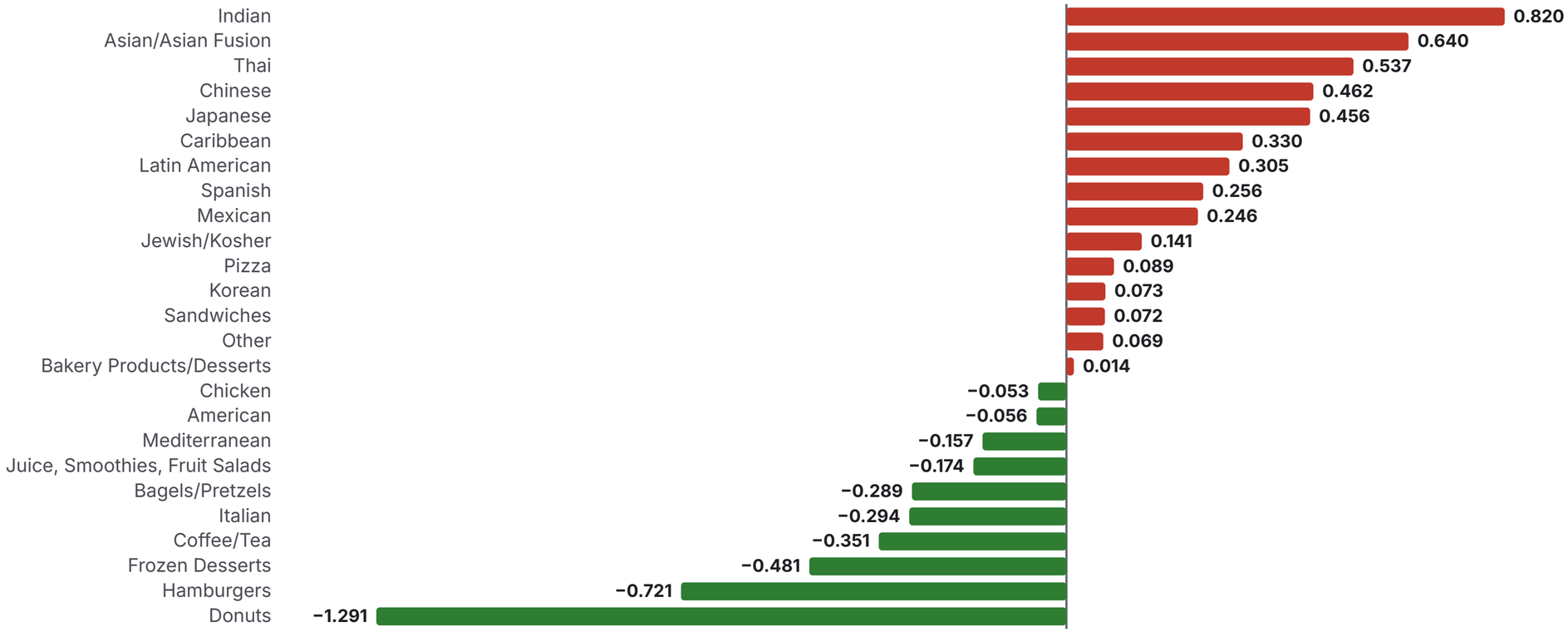
71.0%

recall at 0.5, vs **82.2%** at the chosen **0.45**

Logistic regression on the test years (2025 on). Rows are the true grade, columns the prediction. The notebook's own figure of this matrix is pred_confusion.png.

BACKUP B3 · Q&A

Cuisine pushes the model in both directions



Logistic regression coefficients for grouped cuisine (one-hot, relative to the other levels). Bars to the right push toward B or C; bars to the left push toward A.

BACKUP B4 · Q&A

What the maps cannot say

Inspections are not a random sample

Re-inspections happen because a first visit went badly, so places with more bad first visits get inspected more often. These maps show where inspectors went.

The ≥ 30 floor

A region earns a B or C rate only with at least 30 graded inspections: 178 of 223 ZIPs, 198 of 223 neighborhoods. The rest are grey: not enough inspections to judge.

Boundaries match, almost

Postal ZIPs and Census ZCTAs are cousins: 99.6% of inspections with a ZIP land in a boundary polygon. The neighborhood join placed 94,970 of 95,010 mapped inspections.

A rate describes a place's inspections

A high-rate ZIP describes the inspected activity there, not any single restaurant in it.